



DX-NR01 2.4G Module Technical Manual

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Update Record

Version	Date	Explanation	Author
Version 1.0	2024/01/10	Initial version	LSL
Version 1.1	2024/02/25	New hardware component diagrams	LSL
Version 2.0	2025/03/20	Add module information	YXR

Contact Us

Shenzhen Daxia Longque Technology Co., Ltd.

Email: sales@szdx-smart.com

Telephone: 0755-2997 8125

Website: www.szdx-smart.com

Address: Room 601, Building A1, Huafeng Zhigu, Hangkong Road, Hangcheng Subdistrict, Bao'an District, Shenzhen City

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1. Module Introduction

1.1. Overview

The DX-NR01 is a small-sized 2.4GHz wireless transceiver RF module developed by Shenzhen Daxia Longque Technology Co., Ltd. based on the Si24R1. It features high speed and high stability. The module is equipped with a high-performance PCB antenna and uses an industrial-grade 16MHz crystal oscillator. It operates in the 2.4GHz to 2.525GHz ISM band and is suitable for various IoT applications such as smart home, wireless remote control gesture devices, wireless keyboards and mice, etc. It is an ideal product for IoT applications.

NR01 is a pure RF transceiver module and requires driving by an MCU or a dedicated SPI debugging tool.

1.2. Features

The operating frequency band is 2.4 to 2.525 GHz, with a total of 126 channels, meeting the application requirements of multi-point communication, packet transmission, frequency hopping, etc.

Supports global license-free ISM 2.4GHz band operation

Supported air interface rates: 2Mbps/1Mbps/250Kbps

Industrial-grade crystal oscillator: 16 MHz $\pm$ 10 ppm

Volume and size: 12 $\times$ 19MM

Under ideal conditions, the communication distance can reach up to 230 meters.

Supports power supply from 2.0V to 3.6V

The maximum transmission power can reach up to 7 dBm.

Ultra-low standby power consumption

SPI communication: Connected to the MCU via the SPI interface, with a rate of 0 to 10 Mbps.

☒ Built-in PCB board antenna

1.3. Application

wearable devices

Smart home devices and industrial sensors, etc.

Security systems, positioning systems;

Wireless remote control, unmanned aerial vehicle;

Wireless game remote control;

Medical and health care products;



Wireless voice, wireless headphones;

Children's location and tracking devices;

Logistics tracking, warehouse inspection, electronic tags;

Automotive industry applications.

1.4. Basic Parameters

Table 1: Limit Parameters Table

Main parameters	Performance		Note
	minimum value	maximum value	
Power supply voltage (V)	0	3.6	There is a risk of burning out the module if the voltage exceeds 3.6V.
Blocking power (dBm)	-	10	
Working temperature (°C)	-40	85	industrial grade

Table 2: Working Parameters Table

2. Application Interface

2.1. Module Pin Definition

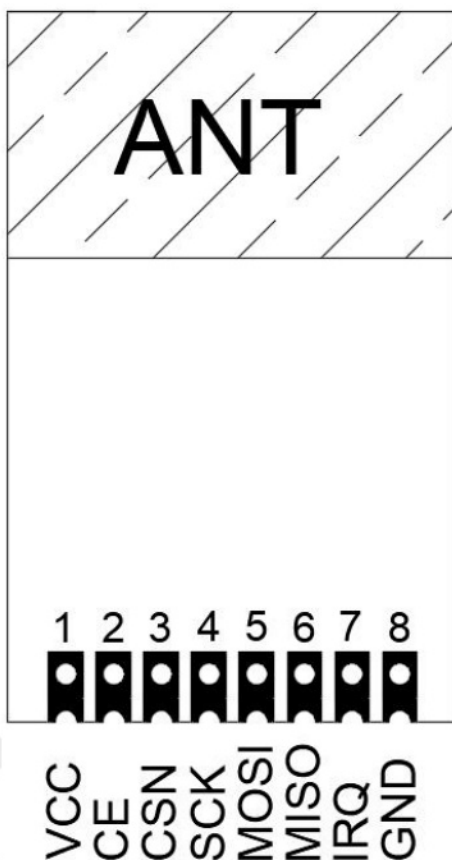


Figure 1: Module Pin Definitions

2.2. Pin definition description

Table 3: Pin Definition Description Table

Pin number	Pin name	Pin functions	Explanation
1	VCC	Power input pins	3.3V (typical value)
2	CE	Module enable control pin	-
3	CSN	Module chip select pin	-
4	SCK	Module SPI bus clock	-

2.3. Power interface design

Table 4: Pin Definition Table for Power Interface

Pin name	Pin number	Description	minimum value	Typical value	maximum value	unit
VCC	1	Module power supply	2.0	3.3	3.6	V
GND	8	ground	-	0	-	V

2.4. Reference connection circuit

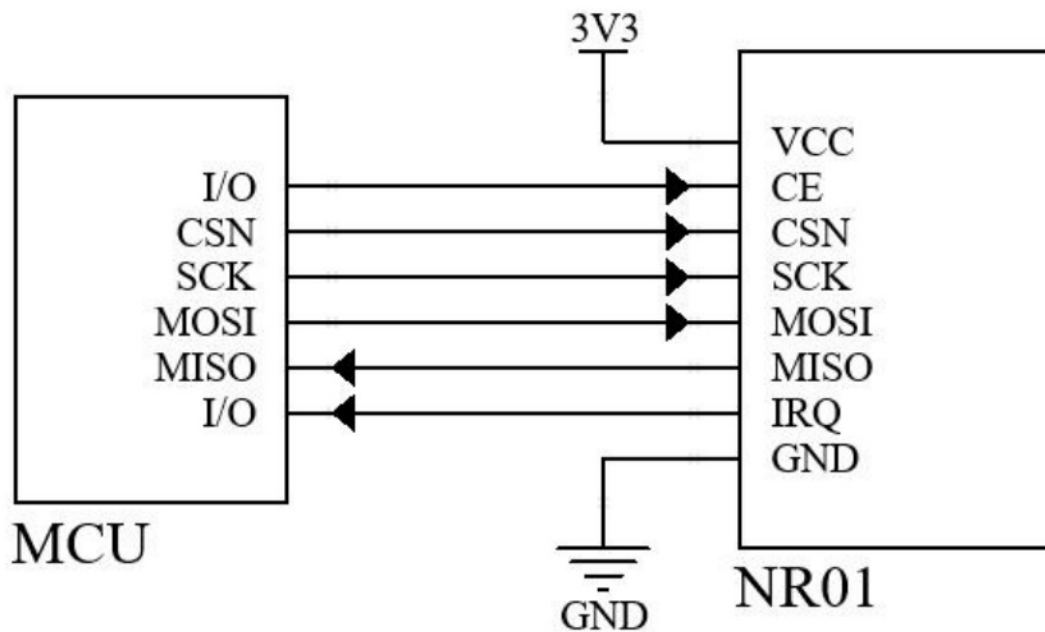


Figure 2: Typical Application Circuit



3. Electrical characteristics, RF characteristics and reliability

3.1. Specification parameters

Table 5: Specification Parameters Table

This image shows a blank sheet of white paper with horizontal ruling lines. At the top, there is a solid gray header bar. A large, light gray watermark is oriented diagonally across the page, featuring the Chinese characters "科技有限" (Technology Limited). The watermark is composed of two parts: "科技" (Technology) and "有限" (Limited), which are slightly offset from each other.

3.2. Electrostatic protection

In module applications, static electricity generated by human body static, charged friction between microelectronic components, etc., discharges to the module through various channels, which may cause certain damage to the module. Therefore, ESD protection should be taken seriously. During the processes of R&D, production assembly and testing, especially in product design, ESD protection measures should be adopted. For instance, anti-static protection should be added at the interface of circuit design and points that are vulnerable to static discharge damage or influence. During production, anti-static gloves should be worn, etc.

Table 6: ESD Tolerance Voltage Table for Module Pins

Test interface	Contact discharge	Air discharge	unit
VCC and GND	+4	+8	kV
Main antenna interface	+2.5	+4	kV



4. Mechanical dimensions and layout suggestions

This section describes the mechanical dimensions of the module. All dimensions are in millimeters; for all dimensions without specified tolerances, the tolerance is ± 0.3 mm.

4.1. Module mechanical ruler

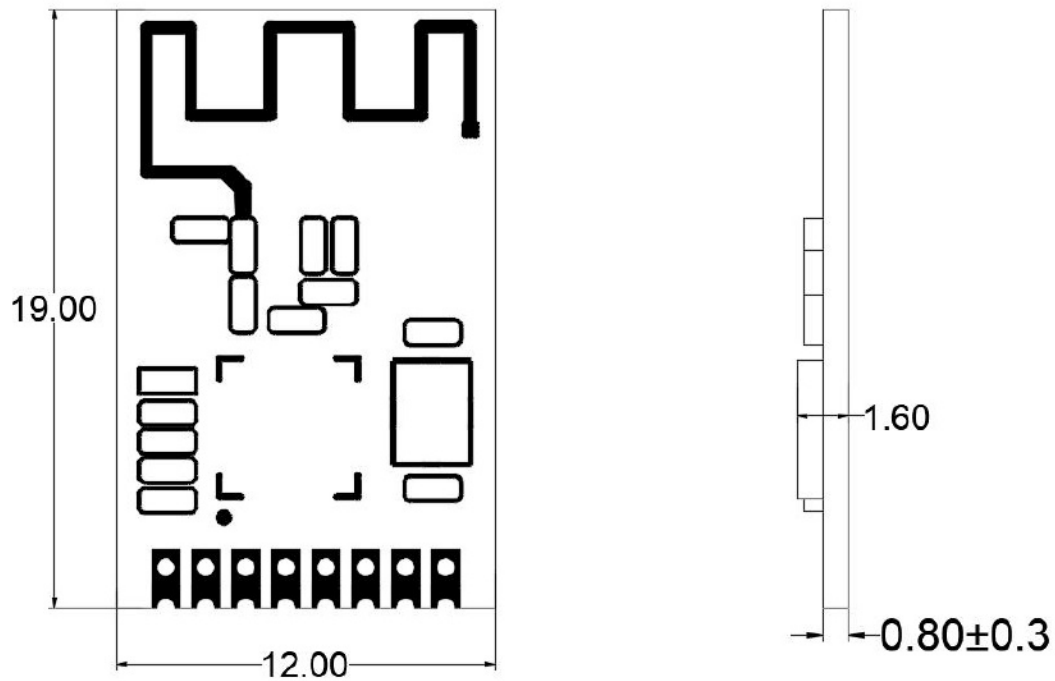


Figure 3: Top and side view dimension drawing of the module

4.2. Recommended packaging

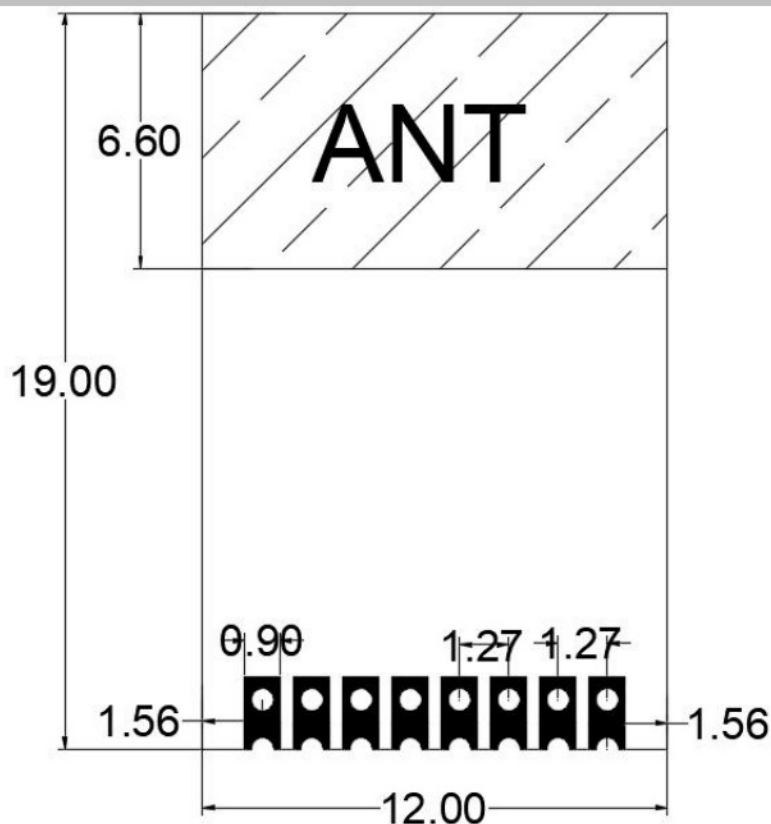


Figure 4: Suggested Encapsulation Size Diagram

4.3. Module top view/bottom view

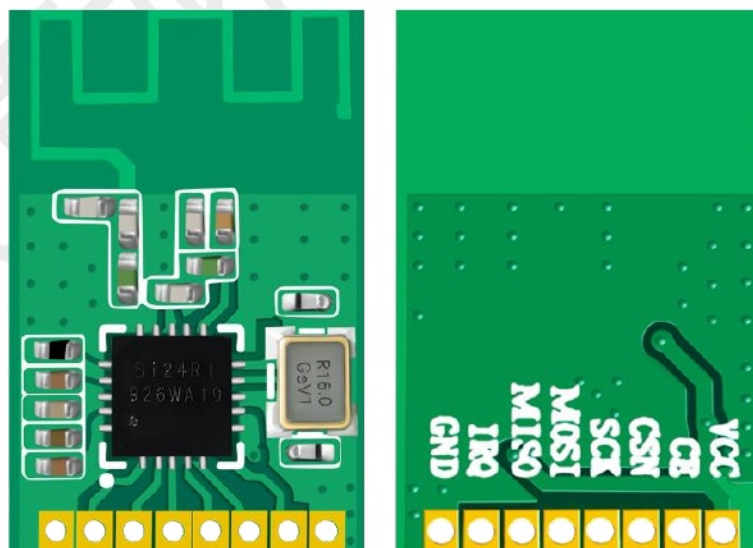


Figure 5: Top view and bottom view of the module



Note

The above picture is for reference only. For the actual appearance and label information of the product, please refer to the physical module.

4.4. Hardware design layout suggestions

The NR01 2.4G module operates in the 2.4G wireless frequency band and uses an onboard antenna. The antenna's Voltage Standing Wave Ratio (VSWR) and efficiency are dependent on the patch position. To minimize the impact of various factors on wireless transmission and reception signals, please note the following points:

The product casing surrounding the Bluetooth module should avoid using metal. When a partial metal casing is used, the antenna part of the module should be kept as far away from the metal part as possible. Internal metal connection wires or metal screws should also be kept away from the antenna part of the module.

- The module antenna part should be placed close to the edge of the carrier board PCB or directly exposed outside the board. It is not allowed to be placed in the middle of the board. There should be at least 5mm of free space in the direction of the antenna, and the area under the antenna on the carrier board should be milled out. Copper plating and wiring are not allowed in the direction parallel to the antenna.

It is recommended to use insulating materials to isolate the module mounting positions on the substrate, for example, by placing a solid piece of silk screen (TopOverLay) at that location.

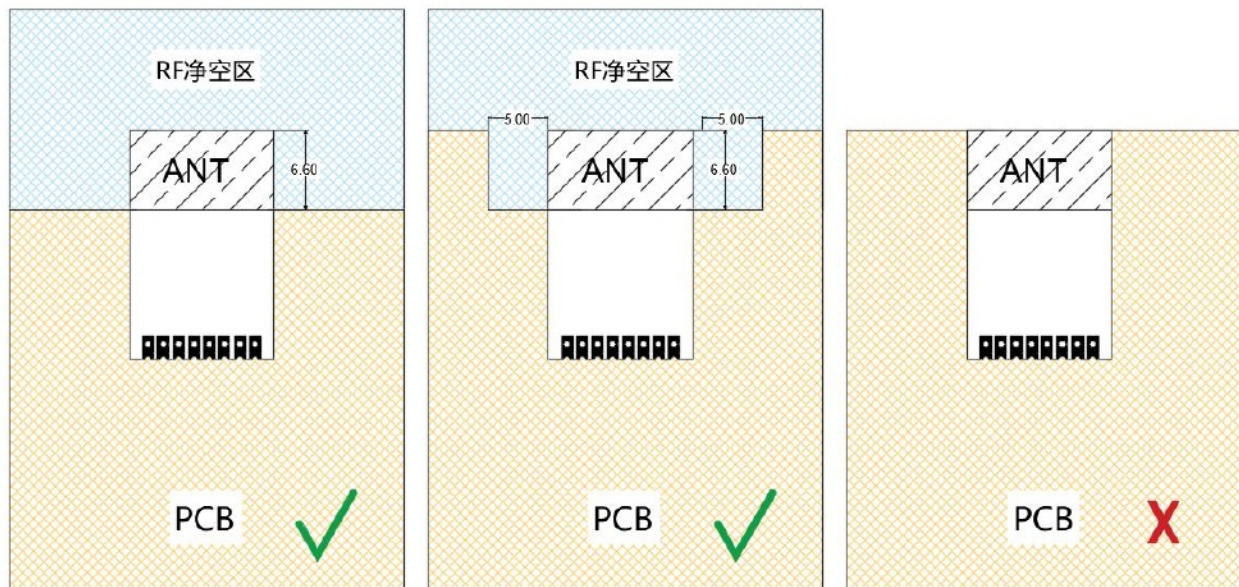


Figure 6: Reference Position for Module Placement

5. Storage, production and packaging

5.1. Storage conditions

The modules are shipped in vacuum-sealed bags. The humidity sensitivity level of the modules is 3 (MSL 3), and their storage must comply with the following conditions:

Recommended storage conditions: temperature $23 \pm 5^{\circ}\text{C}$, and relative humidity 35 - 60%.

Under the recommended storage conditions, the module can be stored in a vacuum-sealed bag for 12 months.

Under workshop conditions of $23 \pm 5^{\circ}\text{C}$ and a relative humidity below 60%, the workshop life of the module after unpacking is 168 hours. Under these conditions, the module can be directly subjected to reflow production or other high-temperature operations. Otherwise, the module needs to be stored in an environment with a relative humidity of less than 10% (e.g., a moisture-proof cabinet) to maintain its dryness.

4. If the module is in the following conditions, pre-baking treatment of the module is required to prevent PCB blistering, cracking and delamination caused by moisture absorption and subsequent high-temperature soldering:

The storage temperature and humidity do not meet the recommended storage conditions.

The module was unpacked but production or storage was not completed in accordance with item 3 above;

Vacuum packaging leakage, loose materials;

Before module repair;

5.2. Module baking process

It needs to be baked at a high temperature of $120 \pm 5^{\circ}\text{C}$ for 8 hours.

Modules that have undergone secondary baking must be welded within 24 hours after baking; otherwise, they still need to be stored in the drying oven.

Note

To prevent and reduce the occurrence of poor welding such as bubbling and delamination of the modules due to moisture, strict control measures should be implemented. It is not recommended to expose the modules to the air for a long time after opening the vacuum packaging.

Before baking, the module should be removed from the package and placed on a heat-resistant container to prevent damage to the plastic tray or reel from high temperatures. Modules that have undergone secondary baking must be welded within 24 hours after baking; otherwise, they should be stored in a drying oven. When unpacking and placing the modules, please pay attention to ESD protection, such as wearing anti-static gloves.

5.3. Reflow soldering

Use a squeegee to print solder paste onto the screen, allowing the solder paste to pass through the openings of the screen and be printed onto the PCB. The force applied by the squeegee should be adjusted appropriately. To ensure the quality of the solder paste printed on the module, it is recommended that the thickness of the steel mesh corresponding to the module's pad area be 0.1 to 0.15mm.

The recommended reflow soldering temperature is 235 to 250 °C, and it must not exceed 250 °C. To prevent damage to the module due to repeated heating, it is strongly recommended

It is recommended that the customer attach the module after the first side of the PCB board has completed reflow soldering. The recommended reflow soldering temperature profile (lead-free SMT reflow soldering) and related parameters are shown in the following chart:

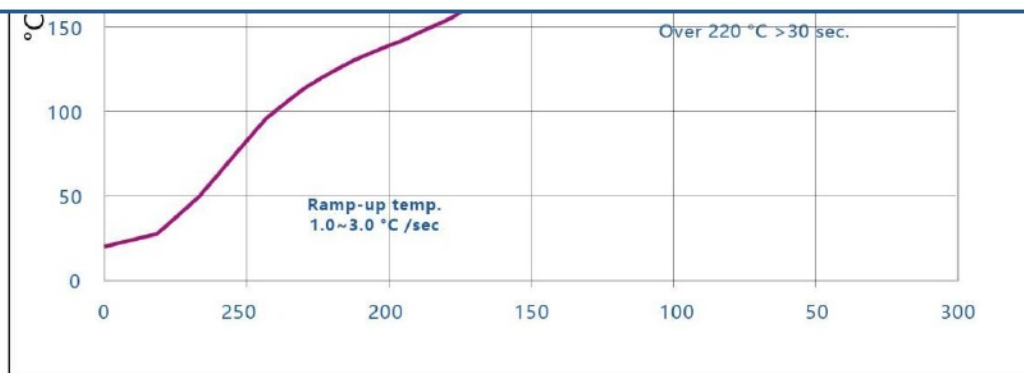


Figure 7: Recommended Reflow Soldering Temperature Profile

Table 7: Recommended Reflow Soldering Temperature

Statistical Name	lower limit	upper limit	unit
The slope 1 (target = 2.0) is between 30.0 and 70.0.	1	3	degrees per second
The slope of 2 (target = 2.0) is between 70.0 and 150.0.	1	3	degrees per second
The slope of 3 (target = -2.8) is between 220.0 and 150.0.	-5	-0.5	degrees per second
Constant temperature time: 110 - 190°C	60	120	second
Reflux time at 220°C	30	65	second
peak temperature	235	250	degree Celsius
The total time at 235°C	10	30	second